

Restaurant Management System - Business Rules

Authentication & Authorization

1. **Login Requirements:** Login must require email, password, and role fields. All fields are mandatory.
2. **Login Validation:** Login must fail if the email does not exist in the system for the specified role.
3. **Password Verification:** Login must fail if the password does not match the stored password for the user.
4. **Login Response:** Login response must return user ID, email, and role. Password and phone must not be included.
5. **Registration Requirements:** Registration must require email, password, and phone number. All fields are mandatory.
6. **Email Format Validation:** Email format must be validated using standard regex pattern during registration and login.
7. **Phone Number Validation:** Phone number must be exactly 10 digits during registration.
8. **Email Uniqueness:** Email must be unique per role during registration. Multiple users can have the same email if they have different roles.
9. **Password Security:** Password must never be returned in any login or registration response.
10. **Role Validation:** Role must strictly be one of: CUSTOMER, WAITER, CHEF, or MANAGER. Role comparison is case-sensitive.

User Management

11. **Manager Privileges:** Only users with MANAGER role can create new staff accounts (WAITER, CHEF).
12. **User Creation:** Managers can view and manage all users except other managers.
13. **Unique Identifiers:** All user IDs must be unique across the system.
14. **Role-Based Access:** Users can only access dashboards and features specific to their assigned role.

Booking Management

15. **Customer Booking:** Only CUSTOMER role users can create table bookings for themselves.

16. **Booking Requirements:** Each booking must include customerId, bookingTime, numberOfPeople, and status.
17. **Future Bookings Only:** Booking time cannot be in the past. Only future dates and times are allowed.
18. **Guest Count Validation:** Number of people in a booking cannot be negative or zero.
19. **Booking Status:** New bookings are created with status “CONFIRMED”. Status can be changed to “CANCELLED”.
20. **Cancellation Irreversibility:** Once a booking status is changed to “CANCELLED”, it cannot be reverted to “CONFIRMED”.
21. **Waiter Booking Access:** Waiters can view all bookings for the current day to prepare for expected guests.
22. **Customer Booking Access:** Customers can only view and manage their own upcoming bookings.

Order Management

23. **Waiter Order Creation:** Only WAITER role users can create new orders.
24. **Order Requirements:** Orders must include tableId, waiterId, and at least one item. TableId cannot be null.
25. **Item Quantity:** Each item in an order must have a quantity greater than or equal to 1. Quantity cannot be zero.
26. **Order Status Flow:** Order status must follow the sequence: PENDING → PREPARING → PREPARED → SERVED.
27. **Initial Order Status:** New orders are automatically created with status “PENDING”.
28. **Chef Status Updates:** Chefs can change order status from PENDING to PREPARING, and from PREPARING to PREPARED.
29. **Waiter Status Updates:** Waiters can change order status from PREPARED to SERVED. Waiters have visibility of all non-served orders.
30. **Order Filtering:** Orders with status “SERVED” are filtered out from waiter’s active order list.